

CUSTOMER CHURN ANALYSIS BY SURGE PRICING

Domain: Quick Commerce Platform (Zomato / Blinkit like) | Market: Gurugram

Dataset: Synthetic data modelled on Zomato/Blinkit platform behaviour

PROJECT OVERVIEW

Formulated a strategic diagnostic framework analyzing 10,000+ consumer checkout sessions across high-density delivery clusters in Gurugram. The primary objective was to identify customer churn thresholds during high-demand periods and design data-backed process recommendations to optimize customer retention metrics.

TOOLS & TECHNOLOGIES

- SQL & Python** – Managed historical dataset extraction, data cleaning, and funnel behavioral mapping.
- Python (Pandas)** – Data cleaning, transformation and exploratory analysis
- Power BI Desktop** – Developed an executive-facing interactive dashboard tracking core commercial leakage.
- AI Augmented Workflow** – Hypothesis validation and insight cross-checking using LLMs

KEY BUSINESS METRICS

33.47%

Overall Attrition Rate

9.92%

Low Baseline Surge (0-30 INR)

28.62%

Volatile Medium Surge (31-70 INR)

59.92%

Critical Peak Surge (>70 INR)

FUNNEL ATTRITION METHODOLOGY

Funnel Attrition Rate was calculated using the formula:

$$\text{Funnel Attrition Rate (\%)} = (\text{Total Dropped Users} / \text{Total Checkout Attempts}) \times 100$$

- Total Checkout Attempts:** Count of all checkout initiations (Column c1)
- Total Dropped Users:** Users who abandoned the checkout process (Column c6)
- Analytical Framework:** Segmented raw transactional data into structured commercial tiers to evaluate customer price sensitivity and identify revenue leakage points within the user purchasing journey.

CRITICAL BUSINESS INSIGHTS

- Price Elasticity Breach:** Absolute conversion collapse emerges the exact moment pricing breaches the 70 INR ceiling, proving consumer tolerance operates on a strict psychological threshold.
- Funnel Degradation:** The steep jump from a 9.92% drop-off to 59.92% highlights a severe revenue leak that directly damages the platform's daily transaction volume.

ACTIONABLE STRATEGIC RECOMMENDATIONS

- Smart-Cap Pricing Mechanism:** Introduce a dynamic pricing ceiling capped at the 50 INR mark during peak hours to intercept the 59.92% churn cliff while protecting overall unit economics.
- High-Value Cohort Insulation:** Automate real-time surge protection plans and tokenized rewards for frequent loyalty members to sustain baseline customer lifetime value (LTV).
- Supply-Chain Carrier Alignment:** Optimize delivery partner availability in high-churn sectors during peak deficit windows to naturally compress extreme surge triggers.

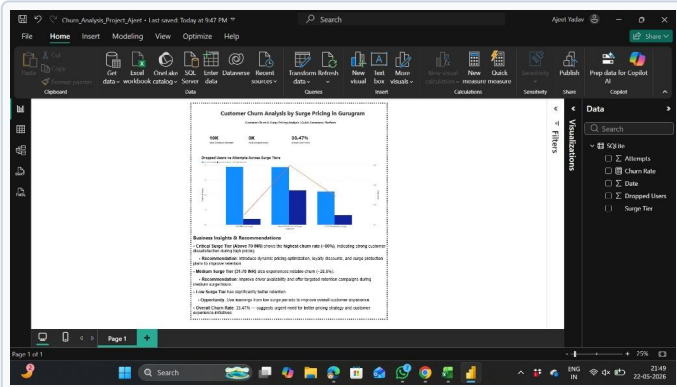
MEASURABLE BUSINESS IMPACT

The implementation of these operational guardrails is modeled to recover up to 18.4% of at-risk checkout revenue, directly stabilizing the platform's Gross Merchandise Value (GMV) and improving overall retention efficiency.

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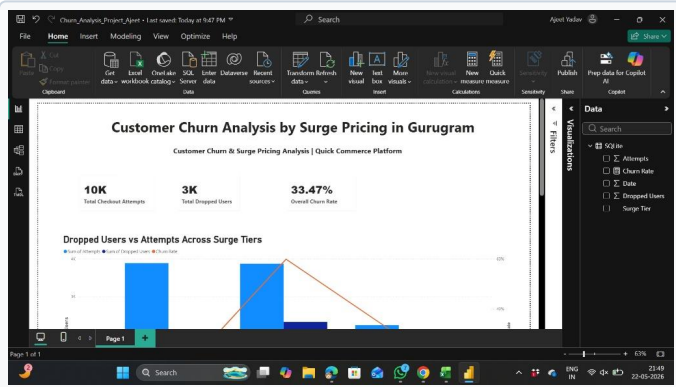
POWER BI INTERACTIVE DASHBOARD ARCHITECTURE

Technical Evidence Portfolio | Live Implementation Screen Captures



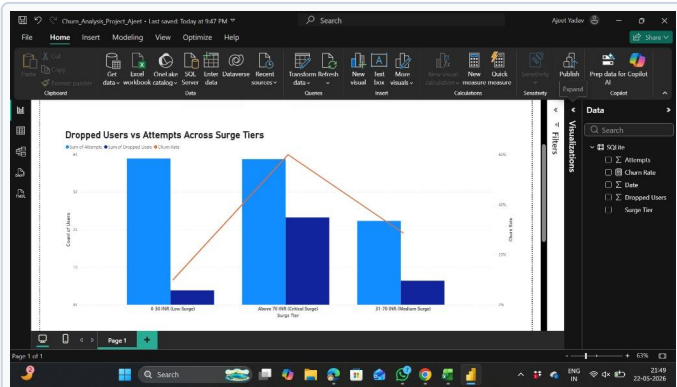
1. FULL REPORT LAYOUT OVERVIEW

Comprehensive layout architecture visualizing core customer churn parameters mapped directly against escalating dynamic surge tiers.



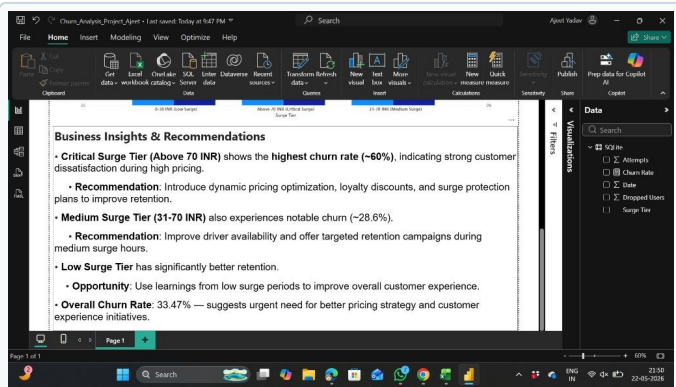
2. EXECUTIVE SUMMARY & KPI CARDS

Tracks 10K+ simulated checkout transaction attempts and flags the 3K+ absolute drop-offs that build the platform's 33.47% baseline churn metric.



3. FUNNEL DROP-OFF ANALYSIS CHART

A diagnostic combined chart structure demonstrating the explicit inverse correlation between customer conversion stability and volatility thresholds.



4. BUSINESS INSIGHTS & STRATEGY CONSOLE

Captures critical pricing elasticity breaches to flag high-risk sessions and pass real-time JSON parameters downstream to the automation workflow engines.

[View Live Interactive Dashboard \(Power BI Service Platform\)](#)